CROW CARRIER — eight synchronous microphone channels / USB data / 12 V operating power

PROTOTYPE — NOT FOR FABRICATION. Outdoor protection and exact assembly qualification remain open.

12 V INPUT / REVERSE PROTECTION / 5 V CONVERSION

J1
VIN12 1
GND 2
CENTER_12V
SLEEVE_GND
PJ-102AH / 12V CENTER POSITIVE

F1
VIN12 1 2
0455001.MRL 1A

D3
1 2
SS34-E3/57T

12V

U10
VIN 1
GND 2
VOUT 3
TSR 1-2450

5V

D2
2 1
VSYS
Diodes Inc 1N5819HW-7-F

D1
LittelFuse SMBJ15A
GND

C1
10uF 50V X7R 10%
GND

12V
C2
10uF 50V X7R 10%
GND

PWR1
VIN12
PWR_FLAG

PWR2
12V
PWR_FLAG

PWR3
GND
PWR_FLAG

PWR4
VSYS
PWR_FLAG

ADC 1: CHANNELS 1-4 / PCM1864 / common 3V3

U1
ADC1_CH1_M 1 VIN1M
ADC1_CH2_M 2 VIN2M
ADC1_CH1_P 3 VIN1P
ADC1_CH2_P 4 VIN2P
5 MIC_BIAS
REF1 6 VREF
GND 7 AGND
3V3 8 DVDD
9 XO
10 XI
LDO1 11 LDO
GND 12 DGND
3V3 13 DVDD
3V3 14 DVDD
MCLK_ADC1 15 SCKI
24 LRCK
17 BCK
18 DOUT
GPO0 19
GPO1 20
MISO 21
SDA 22 SDA
SCL 23 SCL
MCLK 24
GND 25 GND
MOSI 27
VINA1 28 ADC1_CH4_M
VIN3M 29 ADC1_CH3_M
VIN3P 30 ADC1_CH3_P
VIN4P 31 ADC1_CH4_P
PCM1864ADBTR

U1 LOCAL DECOUPLING — separate capacitor positions at each supply pin

REF1
C11
1uF 50V X7R 10% VREF
GND

LDO1
C12
100nF 10V X7R LDO
GND

3V3
C13
100nF 10V X7R AVDD
GND

3V3
C14
10uF 10V X7R AVDD
GND

3V3
C15
100nF 10V X7R DVDD
GND

3V3
C16
10uF 10V X7R DVDD
GND

LDO1
C17
10uF 10V X7R LDO
GND

3V3
C18
100nF 10V X7R IOVDD
GND

ADC 2: CHANNELS 5-8 / PCM1864 / common 3V3

U2
ADC2_CH1_M 1 VIN1M
ADC2_CH2_M 2 VIN2M
ADC2_CH1_P 3 VIN1P
ADC2_CH2_P 4 VIN2P
5 MIC_BIAS
REF2 6 VREF
GND 7 AGND
3V3 8 DVDD
9 XO
10 XI
LDO2 11 LDO
GND 12 DGND
3V3 13 DVDD
3V3 14 DVDD
MCLK_ADC2 15 SCKI
24 LRCK
17 BCK
18 DOUT
GPO0 19
GPO1 20
MISO 21
SDA 22 SDA
SCL 23 SCL
MCLK 24
GND 25 GND
MOSI 27
VINA2 28 ADC2_CH4_M
VIN3M 29 ADC2_CH3_M
VIN3P 30 ADC2_CH3_P
VIN4P 31 ADC2_CH4_P
PCM1864ADBTR

U2 LOCAL DECOUPLING — separate capacitor positions at each supply pin

REF2
C21
1uF 50V X7R 10% VREF
GND

LDO2
C22
100nF 10V X7R LDO
GND

3V3
C23
100nF 10V X7R AVDD
GND

3V3
C24
10uF 10V X7R AVDD
GND

3V3
C25
100nF 10V X7R DVDD
GND

3V3
C26
10uF 10V X7R DVDD
GND

LDO2
C27
10uF 10V X7R LDO
GND

3V3
C28
100nF 10V X7R IOVDD
GND

U3 ASSEMBLY: DEPOPULATE ON-MODULE D1 only: retain R10/R1 VBUS sense. USB DATA ONLY.

U3
SDA 1 GP00
SCL 2 GP01
GND 3 GND
TDM 4 GP02
BCK 5 GP03
LRCK 6 GP04
BUZZ_GPIO1 7 GP05
GND 8 GND
BUZZ_GPIO2 9 GP06
BUZZ_GPIO3 10 GP07
BUZZ_GPIO4 11 GP08
BUZZ_GPIO5 12 GP09
GND 13 GND
BUZZ_GPIO6 14 GP010
BUZZ_GPIO7 15 GP011
BUZZ_GPIO8 16 GP012
17 GP013
GND 18 GND
19 GP014
20 GP015
GP00 40
GP001 39
GND 38
GP002 37
GP003 36
GP004 35
GND 34
GP005 33
GP006 32
GP007 31
AGND 30
GP020 29
ADC_VREF 28
3V3 27
3V3_EN 26
GND 25
VSYS 24
VBUS 23
Pico SC0915 MOD: on-module D1 DEPOPULATED

3V3
R1
4.7k 1%
SDA

3V3
R2
4.7k 1%
SCL

SHARED 12.288 MHz MCLK

X1
3V3 1
GND 2
OE 3
GND 4
OUT
VDD
MCLK_SRC
3V3
CB3LV-3I-12M2880000

3V3
C4
100nF 10V X7R
GND

R3
MCLK_SRC 1 2 MCLK_FAN
68R 1% 0.1W 25V

R4
1 2 MCLK_ADC1
56R 1% 0.1W 25V

R5
1 2 MCLK_ADC2
56R 1% 0.1W 25V

POD 1 — differential audio + power + active-low calibration command

J101
P1_A_P 1 contact_1
P1_A_N 2 contact_2
P1_12V 3 contact_3
P1_12V 4 contact_4
contact_5 5 GND
contact_6 6 GND
contact_7 7 P1_BUZZ_N
contact_8 8 GND
RJHSE-5080

D501
P1_A_P 1
GND 2
P1_A_N 3
5 P1_BUZZ_N
SP0504BAHTG

POD 2 — differential audio + power + active-low calibration command

J102
P2_A_P 1 contact_1
P2_A_N 2 contact_2
P2_12V 3 contact_3
P2_12V 4 contact_4
contact_5 5 GND
contact_6 6 GND
contact_7 7 P2_BUZZ_N
contact_8 8 GND
RJHSE-5080

D502
P2_A_P 1
GND 2
P2_A_N 3
5 P2_BUZZ_N
SP0504BAHTG

POD 3 — differential audio + power + active-low calibration command

J103
P3_A_P 1 contact_1
P3_A_N 2 contact_2
P3_12V 3 contact_3
P3_12V 4 contact_4
contact_5 5 GND
contact_6 6 GND
contact_7 7 P3_BUZZ_N
contact_8 8 GND
RJHSE-5080

D503
P3_A_P 1
GND 2
P3_A_N 3
5 P3_BUZZ_N
SP0504BAHTG

POD 4 — differential audio + power + active-low calibration command

J104
P4_A_P 1 contact_1
P4_A_N 2 contact_2
P4_12V 3 contact_3
P4_12V 4 contact_4
contact_5 5 GND
contact_6 6 GND
contact_7 7 P4_BUZZ_N
contact_8 8 GND
RJHSE-5080

D504
P4_A_P 1
GND 2
P4_A_N 3
5 P4_BUZZ_N
SP0504BAHTG

POD 5 — differential audio + power + active-low calibration command

J105
P5_A_P 1 contact_1
P5_A_N 2 contact_2
P5_12V 3 contact_3
P5_12V 4 contact_4
contact_5 5 GND
contact_6 6 GND
contact_7 7 P5_BUZZ_N
contact_8 8 GND
RJHSE-5080

D505
P5_A_P 1
GND 2
P5_A_N 3
5 P5_BUZZ_N
SP0504BAHTG

POD 6 — differential audio + power + active-low calibration command

J106
P6_A_P 1 contact_1
P6_A_N 2 contact_2
P6_12V 3 contact_3
P6_12V 4 contact_4
contact_5 5 GND
contact_6 6 GND
contact_7 7 P6_BUZZ_N
contact_8 8 GND
RJHSE-5080

D506
P6_A_P 1
GND 2
P6_A_N 3
5 P6_BUZZ_N
SP0504BAHTG

POD 7 — differential audio + power + active-low calibration command

J107
P7_A_P 1 contact_1
P7_A_N 2 contact_2
P7_12V 3 contact_3
P7_12V 4 contact_4
contact_5 5 GND
contact_6 6 GND
contact_7 7 P7_BUZZ_N
contact_8 8 GND
RJHSE-5080

D507
P7_A_P 1
GND 2
P7_A_N 3
5 P7_BUZZ_N
SP0504BAHTG

POD 8 — differential audio + power + active-low calibration command

J108
P8_A_P 1 contact_1
P8_A_N 2 contact_2
P8_12V 3 contact_3
P8_12V 4 contact_4
contact_5 5 GND
contact_6 6 GND
contact_7 7 P8_BUZZ_N
contact_8 8 GND
RJHSE-5080

D508
P8_A_P 1
GND 2
P8_A_N 3
5 P8_BUZZ_N
SP0504BAHTG

C102
P1_A_P 1 2
2.2uF 25V X7R 20%
P1_CP
R102
1 2
2.7k 1%
ADC1_CH1_P
D102
BZT52C3V9-7-F
C301
2.2nF 50V C0G 5%
GND

C103
P1_A_M 1 2
2.2uF 25V X7R 20%
P1_CM
R103
1 2
2.7k 1%
ADC1_CH1_M
D103
BZT52C3V9-7-F
C401
2.2nF 50V C0G 5%
GND

C104
P2_A_P 1 2
2.2uF 25V X7R 20%
P2_CP
R104
1 2
2.7k 1%
ADC1_CH2_P
D104
BZT52C3V9-7-F
C302
2.2nF 50V C0G 5%
GND

C105
P2_A_M 1 2
2.2uF 25V X7R 20%
P2_CM
R105
1 2
2.7k 1%
ADC1_CH2_M
D105
BZT52C3V9-7-F
C402
2.2nF 50V C0G 5%
GND

C106
P3_A_P 1 2
2.2uF 25V X7R 20%
P3_CP
R106
1 2
2.7k 1%
ADC1_CH3_P
D106
BZT52C3V9-7-F
C303
2.2nF 50V C0G 5%
GND

C107
P3_A_M 1 2
2.2uF 25V X7R 20%
P3_CM
R107
1 2
2.7k 1%
ADC1_CH3_M
D107
BZT52C3V9-7-F
C403
2.2nF 50V C0G 5%
GND

C108
P4_A_P 1 2
2.2uF 25V X7R 20%
P4_CP
R108
1 2
2.7k 1%
ADC1_CH4_P
D108
BZT52C3V9-7-F
C304
2.2nF 50V C0G 5%
GND

C109
P4_A_M 1 2
2.2uF 25V X7R 20%
P4_CM
R109
1 2
2.7k 1%
ADC1_CH4_M
D109
BZT52C3V9-7-F
C404
2.2nF 50V C0G 5%
GND

C110
P5_A_P 1 2
2.2uF 25V X7R 20%
P5_CP
R110
1 2
2.7k 1%
ADC2_CH1_P
D110
BZT52C3V9-7-F
C305
2.2nF 50V C0G 5%
GND

C111
P5_A_M 1 2
2.2uF 25V X7R 20%
P5_CM
R111
1 2
2.7k 1%
ADC2_CH1_M
D111
BZT52C3V9-7-F
C405
2.2nF 50V C0G 5%
GND

C112
P6_A_P 1 2
2.2uF 25V X7R 20%
P6_CP
R112
1 2
2.7k 1%
ADC2_CH2_P
D112
BZT52C3V9-7-F
C306
2.2nF 50V C0G 5%
GND

C113
P6_A_M 1 2
2.2uF 25V X7R 20%
P6_CM
R113
1 2
2.7k 1%
ADC2_CH2_M
D113
BZT52C3V9-7-F
C406
2.2nF 50V C0G 5%
GND

C114
P7_A_P 1 2
2.2uF 25V X7R 20%
P7_CP
R114
1 2
2.7k 1%
ADC2_CH3_P
D114
BZT52C3V9-7-F
C307
2.2nF 50V C0G 5%
GND

C115
P7_A_M 1 2
2.2uF 25V X7R 20%
P7_CM
R115
1 2
2.7k 1%
ADC2_CH3_M
D115
BZT52C3V9-7-F
C407
2.2nF 50V C0G 5%
GND

C116
P8_A_P 1 2
2.2uF 25V X7R 20%
P8_CP
R116
1 2
2.7k 1%
ADC2_CH4_P
D116
BZT52C3V9-7-F
C308
2.2nF 50V C0G 5%
GND

C117
P8_A_M 1 2
2.2uF 25V X7R 20%
P8_CM
R117
1 2
2.7k 1%
ADC2_CH4_M
D117
BZT52C3V9-7-F
C408
2.2nF 50V C0G 5%
GND

BUZZ_GPIO1
R301
100k 1%
GND
Q101
P1_BUZZ_N
BUZZ_GPIO1
DMN3023L-7
GND
F101
12V 1 2 P1_12V
1206L010/30NR

BUZZ_GPIO2
R302
100k 1%
GND
Q102
P2_BUZZ_N
BUZZ_GPIO2
DMN3023L-7
GND
F102
12V 1 2 P2_12V
1206L010/30NR

BUZZ_GPIO3
R303
100k 1%
GND
Q103
P3_BUZZ_N
BUZZ_GPIO3
DMN3023L-7
GND
F103
12V 1 2 P3_12V
1206L010/30NR

BUZZ_GPIO4
R304
100k 1%
GND
Q104
P4_BUZZ_N
BUZZ_GPIO4
DMN3023L-7
GND
F104
12V 1 2 P4_12V
1206L010/30NR

BUZZ_GPIO5
R305
100k 1%
GND
Q105
P5_BUZZ_N
BUZZ_GPIO5
DMN3023L-7
GND
F105
12V 1 2 P5_12V
1206L010/30NR

BUZZ_GPIO6
R306
100k 1%
GND
Q106
P6_BUZZ_N
BUZZ_GPIO6
DMN3023L-7
GND
F106
12V 1 2 P6_12V
1206L010/30NR

BUZZ_GPIO7
R307
100k 1%
GND
Q107
P7_BUZZ_N
BUZZ_GPIO7
DMN3023L-7
GND
F107
12V 1 2 P7_12V
1206L010/30NR

BUZZ_GPIO8
R308
100k 1%
GND
Q108
P8_BUZZ_N
BUZZ_GPIO8
DMN3023L-7
GND
F108
12V 1 2 P8_12V
1206L010/30NR

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